

PENDYALA SIRI

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PROFESSIONAL SUMMARY

AI & Data Science graduate (2026) with 10+ months of production experience on data and ML platforms: **PostgreSQL** administration and schema migrations, **batch and asynchronous data pipelines** (Celery, RabbitMQ), and running **ML models in production** (NVIDIA Triton, Docker, Kubernetes, AWS) under strict security and change controls. Strong in **SQL and Python**, with hands-on data modeling, query optimization, automated testing and **CI/CD** (GitHub Actions). First author of a published paper on LLM hallucinations proposing a RAG + RLHF system, and currently building a multi-agent **GenAI workflow** in LangGraph. Uses **AI tools** (Claude, ChatGPT, Codex) daily for coding, testing, documentation and troubleshooting.

TECHNICAL SKILLS

SQL & Data: SQL, PostgreSQL, Data modeling, SQL query optimization & indexing, Schema migrations (Alembic), MongoDB, SQLite, ETL / ELT concepts, Dataset curation & preprocessing, Curated reporting datasets

Pipelines & Processing: Python, Celery, RabbitMQ (message queues), Batch & asynchronous job pipelines, Scheduled jobs, Real-time streaming (WebSockets), MinIO (S3-compatible object storage)

MLOps & Deployment: Model serving (NVIDIA Triton, FastAPI), Docker, K3s (Kubernetes), Versioned releases & rollbacks, Nexus Repository, Latency profiling, Model quantization

CI/CD, DevOps & Cloud: GitHub Actions (CI/CD), Git, AWS, Linux, HAProxy, Bash

Data Quality & Testing: pytest, Unit / integration / regression testing, Edge-case validation, Playwright, Root-cause analysis

AI, ML & GenAI: PyTorch, scikit-learn, Computer vision, LLMs, RAG, Prompt engineering, OpenAI API, Gemini API, LangGraph, Qdrant, Claude, ChatGPT, Codex

Languages: Python, SQL, JavaScript, TypeScript, Java, R, Bash

PROFESSIONAL EXPERIENCE

Junior Software Engineer | Jul 2026 – Present

Center for Visual Information Technology (CVIT), IIIT Hyderabad · Hyderabad, India

- Administer PostgreSQL databases and Alembic schema migrations for a production AI platform (New Anubhava Mantapa), and configure HAProxy for database and application traffic, improving production stability and load distribution.
- Manage batch and asynchronous processing pipelines with Celery and RabbitMQ that move data between application services, NVIDIA Triton model inference, MinIO object storage and a Qdrant face-recognition vector search pipeline, supporting reliable high-volume background processing.
- Support ML models and services in production inside a secure air-gapped network under strict access, governance and change controls.
- Own versioned releases, configuration updates and rollback readiness, control package distribution through Nexus Repository, and maintain architecture and troubleshooting documentation.
- Diagnose and resolve production issues across containerized services (Docker, K3s) through root-cause analysis, coordinating fixes with developers and testers.

AI Engineer Intern | Apr 2026 – Jul 2026

Zeto.studio · Hyderabad, India

- Engineered a low-latency real-time streaming pipeline (RTF 0.6, TTFB 1.2s) on a WebSocket-based full-duplex speech-to-text / text-to-speech architecture with sentence-level chunk delivery, containerized with Docker and deployed on AWS.
- Reduced per-request latency and output errors by profiling bottlenecks and applying dynamic batching, model quantization and voice activity detection (VAD).

Junior Software Engineer Intern | Nov 2025 – Apr 2026

Center for Visual Information Technology (CVIT), IIIT Hyderabad · Hyderabad, India

- Deployed an AI Text-to-Speech model (INDIC F5) as a FastAPI REST service for real-time inference, and built computer vision pipelines for automated background removal.
- Built pytest unit and integration suites covering edge-case failures for regression coverage, and wrote architecture, functional specification and troubleshooting documentation.

PROJECTS

InsightHub — Business Intelligence Platform | Python, FastAPI, PostgreSQL, SQL, Apache Airflow

- Built a reporting platform that produces curated datasets using multi-table SQL joins, indexed queries and parallel aggregations, with role-based access control for secure multi-user reporting.
- Added an Apache Airflow ETL pipeline that extracts public data, cleans and transforms it, runs data quality checks (schema, null and row-count validation) and loads curated tables into PostgreSQL, with run history for lineage.

KALANETRA — Mohiniyattam Mudra Detection | Python, MediaPipe, scikit-learn

- Built an end-to-end data and feature pipeline: personally collected a 300-image dataset across 24 mudra classes, then extracted MediaPipe hand keypoints to turn raw images into a structured feature dataset for model training.
- Trained a landmark-based classifier on the extracted features, reaching 85% accuracy under limited training data.

VeriAgent — Multi-Agent GenAI Workflow | LangGraph, LangChain, Gemini API, Qdrant, FastAPI

- Built an agentic workflow (planner, retriever, writer, critic) over Qdrant-indexed documents, with an evaluation set that measures answer quality and latency.

EDUCATION & RESEARCH

B.E. in Artificial Intelligence and Data Science, Methodist College of Engineering and Technology, Hyderabad — Nov 2022 – June 2026 | CGPA: 8.7 / 10 | Coursework: Cloud Computing (AWS)

First Author — “Detecting Hallucinations in Large Language Models”, GRENZE International Journal of Engineering and Technology, 2025. Proposed a combined RAG + RLHF system to address hallucinations in LLM output.

LEADERSHIP

Divisional President (2023–24) and VP – Education (2022–23), Street Cause MCET; Director of Education, Hyderabad Youth Assembly; Student Ambassador, WE Hub